

SIMATS ENGINEERING
CO2-ASSESSMENT TOOL 2

COURSE NAME: Design and Analysis of Algorithms

COURSE CODE: CSA0613

Name : C.Himaja

Reg.No : 192373035

Q11. A banking system stores daily transaction amounts in a list where new entries are appended randomly, causing disorder. The system needs to sort data periodically for reporting.

- a) Analyze how Selection Sort and Bubble Sort would work in this scenario.**
- b) Compare the number of comparisons and swaps required by both methods.**
- c) Recommend the most suitable sorting technique with proper justification.**

Source code:

```
#include <stdio.h>
```

```
void selectionSort(int arr[], int n) {  
    int i, j, min, temp;  
    for(i = 0; i < n - 1; i++) {  
        min = i;  
        for(j = i + 1; j < n; j++) {  
            if(arr[j] < arr[min])  
                min = j;  
        }  
        if(min != i) {  
            temp = arr[i];  
            arr[i] = arr[min];  
            arr[min] = temp;  
        }  
    }  
}
```

```
    }  
}
```

```
void bubbleSort(int arr[], int n) {  
    int i, j, temp;  
    for(i = 0; i < n - 1; i++) {  
        for(j = 0; j < n - i - 1; j++) {  
            if(arr[j] > arr[j + 1]) {  
                temp = arr[j];  
                arr[j] = arr[j + 1];  
                arr[j + 1] = temp;  
            }  
        }  
    }  
}
```

```
void printArray(int arr[], int n) {  
    int i;  
    for(i = 0; i < n; i++)  
        printf("%d ", arr[i]);  
    printf("\n");  
}
```

```
int main() {  
    int n, i;  
  
    printf("Enter number of transactions: ");  
    scanf("%d", &n);  
  
    int a[n], b[n];  
  
    printf("Enter transaction amounts:\n");
```

```
for(i = 0; i < n; i++) {  
    scanf("%d", &a[i]);  
    b[i] = a[i];  
}  
  
selectionSort(a, n);  
printf("\nAfter Selection Sort:\n");  
printArray(a, n);  
  
bubbleSort(b, n);  
printf("\nAfter Bubble Sort:\n");  
printArray(b, n);  
  
return 0;  
}
```

Sample Input

Enter number of transactions: 6

Enter transaction amounts:

4500

1200

3800

2500

5100

1900

Sample Output

After Selection Sort:

1200 1900 2500 3800 4500 5100

After Bubble Sort:

1200 1900 2500 3800 4500 5100

Time Complexity

Algorithm	Best Case	Average Case	Worst Case
-----------	-----------	--------------	------------

Selection Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
----------------	----------	----------	----------

Bubble Sort	$O(n^2)$	$O(n^2)$	$O(n^2)$
-------------	----------	----------	----------

Space Complexity

Both algorithms use **$O(1)$** auxiliary space.

Result

Both Selection Sort and Bubble Sort correctly sort the banking transaction amounts in ascending order. Selection Sort performs fewer swaps, while Bubble Sort performs many more swaps during sorting.